

The treatment of advanced melanoma: Current approaches and new challenges

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ABSTRACT

In recent years, advances in melanoma treatment have renewed patient hope. This comprehensive review emphasizes the evolving treatment landscape, particularly highlighting first-line strategies and the interplay between immune-checkpoint inhibitors (ICIs) and targeted therapies. Ipilimumab plus nivolumab has achieved the best median overall survival, exceeding 70 months. However, the introduction of new ICIs, like relatlimab, has added complexity to first-line therapy decisions. Our aim is to guide clinicians in making personalized treatment decisions. Various features, including brain metastases, PD-L1 expression, BRAF mutation, performance status, and prior adjuvant therapy, significantly impact the direction of advanced melanoma treatment. We also provide the latest insights into the treatment of rare melanoma subtypes, such as uveal melanoma, where tebentafusp has shown promising improvements in overall survival for metastatic uveal melanoma patients. This review provides invaluable insights for clinicians, enabling informed treatment choices and deepening our understanding of the multifaceted challenges associated with advanced melanoma management.

1. Introduction

Melanoma has historically been a formidable challenge in oncology, particularly in its advanced stages. The landscape of melanoma treatment has undergone a dramatic transformation in the past decade, largely due to the advent of immune-checkpoint inhibitors (ICIs) and targeted therapies (Garbe et al., 2011; Hodi et al., 2022).

Advanced melanoma, defined as unresectable or metastatic disease, has been historically associated with poor prognosis (Long et al., 2023). The introduction of ICIs targeting CTLA-4, PD-1, and PD-L1, as well as BRAF and MEK inhibitors, has significantly improved outcomes. Despite these advances, challenges remain, including the management of brain metastases, rare melanoma subtypes like uveal and mucosal melanoma, and the emergence of resistance to current therapies.

This review synthesizes the latest evidence on the effectiveness of various treatment modalities, with a focus on ICIs and BRAF/MEK inhibitors. It highlights the pivotal role of combining ICIs, the impact of

BRAF mutations on treatment decisions, and the importance of considering individual patient factors such as tumor burden, performance status, and prior therapies. The management of brain metastases and rare melanoma subtypes, which require tailored approaches, is also discussed in detail.

In addition to the comprehensive overview of current treatment approaches for advanced melanoma, this review also provides practical treatment algorithms. These algorithms were designed to guide clinicians through the intricacies of decision-making in the management of advanced melanoma. Considering various clinical scenarios, including patient characteristics, tumour biology, and prior treatments received.

These treatment algorithms are particularly valuable in navigating the complex landscape of advanced melanoma, where personalized treatment is key. They offer a structured approach to selecting the most appropriate therapeutic strategies based on factors such as the presence or absence of BRAF mutations, the presence of brain metastases and patient performance status.

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By incorporating these algorithms into clinical practice, oncologists and healthcare professionals can make more informed decisions, optimize treatment outcomes, and navigate the challenges of treating advanced melanoma more effectively. The algorithms serve as a practical tool to complement the detailed information provided in the review, ensuring a comprehensive approach to the management of this challenging condition.

2. Genomic subclassification of melanoma

Melanoma, traditionally perceived as a uniform malignancy with a poor prognosis, has emerged as a collection of diverse subtypes, each characterized by distinct clinical presentations, histopathological and genetic features (Akbanı et al., 2015). The last decades have unravelled the heterogeneity of melanoma, prompting a paradigm shift in its classification and management.

2.1. Cutaneous melanoma

Cutaneous melanoma, the most prevalent form, has been the primary focus of clinical research and is further divided based on sun exposure into chronic sun-damaged (CSD), non-CSD, and acral melanomas (Long et al., 2023; Akbanı et al., 2015; Romano et al., 2011). The WHO has categorized cutaneous melanoma based on molecular characteristics, into nine distinct subtypes (Akbanı et al., 2015). This molecular classification hinges on specific genetic alterations and sun exposure patterns, giving rise to subtypes with unique clinical and histopathological profiles.

The two principal pathways to cutaneous melanoma genesis are determined by the degree of cumulative sun damage: superficial spreading melanoma for low damage, and lentigo maligna melanoma for high damage (Akbanı et al., 2015). Superficial spreading melanoma, often linked to a precursor naevus, is predominantly seen in younger adults and is characterized by BRAFV600E mutations (Long et al., 2023). Lentigo maligna melanoma, on the other hand, lacks these mutations but may feature alterations in NF1 or NRAS and arises in sun-damaged skin of older individuals (Long et al., 2023). Desmoplastic melanoma stands out as a rare subtype, commonly presenting in sun-damaged skin of the elderly and challenging to diagnose. It is marked by a high tumour mutation burden, correlating with reduced sentinel lymph node involvement and enhanced responsiveness to immunotherapy (Romano et al., 2011).

Acral melanoma is characterized by unique genomic aberrations compared to melanomas that arise from non-chronically sun-damaged skin (Akbanı et al., 2015). While non-chronically sun-damaged melanomas frequently have activating mutations in BRAF and an increased frequency of NRAS mutations, acral melanomas typically do not show these oncogenic mutations. Instead, acral melanomas are associated with increased copy numbers of cyclin-dependent kinase 4 (CDK4) and cyclin D1 (CCND1), which are components downstream in the pathway (Romano et al., 2011).

2.2. Non-cutaneous melanoma

Beyond cutaneous, melanoma presents in less common but complex forms such as mucosal and uveal melanomas. Mucosal melanomas, arising from melanocytes in mucosal tissues, present challenges due to their occult locations and aggressive nature. Mucosal melanomas are less frequently associated with UV signature mutations, which are common in other melanoma types. This lack of UV signature suggests that mucosal melanomas may arise through different mutational processes, potentially involving more complex genomic rearrangements (Romano et al., 2011).

Key molecular features of mucosal melanoma include a higher frequency of KIT mutations and amplifications, as well as amplifications of the RTKs PDGFRA and KDR (VEGFR2). These findings are significant

because they can offer potential targets for treatment. For example, KIT mutations may respond to targeted therapies such as imatinib (Hodi et al., 2013). Moreover, the amplifications of PDGFRA and KDR suggest that therapies targeting these receptors could be effective for patients with mucosal melanoma.

Furthermore, mucosal melanomas exhibit unique molecular alterations that are less common in other melanoma subtypes, such as co-amplified RTKs and mutations in genes like SF3B1 (Romano et al., 2011). These genetic aberrations could provide avenues for novel therapeutic strategies. The genomic features observed in mucosal melanomas underline the importance of considering the unique molecular landscape of this subtype when devising treatment plans and therapeutic targets. Uveal melanoma, originating from melanocytes within the vascular layer of the eye, distinguishes itself through GNAQ and GNA11 mutations and lacks the characteristic BRAF mutations commonly seen in cutaneous melanoma (Romano et al., 2011).

The understanding of the specific genetic mutations and pathways involved in different melanoma subtypes, can help clinicians to better tailor treatments to individual patients. This personalized approach is particularly crucial in the context of rare melanoma forms, which often require specialized management strategies.

Furthermore, our review highlights the importance of ongoing research and clinical trials in expanding the therapeutic arsenal for melanoma. With advancements in genomic profiling and molecular biology, new targets for therapy are continually being identified, offering hope for more effective treatment options in the future, even in the most challenging subtypes. The management of melanoma has entered a new era of precision medicine, where the understanding of genetic and molecular pathways opens the way for more informed and effective clinical decisions.

3. Immune-checkpoint inhibitors for the treatment of advanced melanoma

With the introduction of ICIs and targeted therapies (BRAF/MEK inhibitors), the treatment of advanced (unresectable or metastatic) cutaneous melanoma has been revolutionized. In fact, until the introduction of these agents, the median survival (mOS) of patients with advanced melanoma was about 6 months (Garbe et al., 2011). To date, 6-year mOS has been reached with the combination of ICIs ipilimumab/nivolumab (Hodi et al., 2022).

This revolution began with the approval by the U.S. Food and Drug Administration (FDA) of ipilimumab, an inhibitor of Cytotoxic T-Lymphocyte Antigen 4 (CTLA-4). Ipilimumab inhibits a key checkpoint receptor involved in the inhibition of T-lymphocyte activation (Hodi et al., 2010). Ipilimumab was the first agent to be associated with a survival benefit in advanced melanoma, despite a low rate of responses (10%) and a median progression-free survival (PFS) of only 3 months (Hodi et al., 2010). Long-term follow-up demonstrated the true efficacy and effectiveness of ipilimumab in at least a subset of patients with advanced melanoma. In a pooled analysis of long-term survival data from studies with a follow-up of up to 10 years, among 1.861 patients, mOS was 11.4 months, but the survival curve began to plateau around year 3. Three-year survival rates were 26%, and 20% for treatment-naïve and previously treated patients, respectively. Similar results were observed in patients receiving treatment in the expanded access program (Schadendorf et al., 2015).

Yet, ipilimumab as monotherapy is no longer considered a first-line therapy today. In fact, in a direct comparison, the Programmed cell Death-1 (PD-1) inhibitor pembrolizumab at a dose of 10 mg/kg once every 2 weeks and once every 3 weeks, more than doubled both median survival (about 32 months versus 16 months) and PFS compared to ipilimumab (about 8 months versus 3 months), while also showing a slightly more favorable toxicity profile, with a rate of grade 3–4 immune-related adverse events (irAEs) of about 15% (Robert et al., 2015).

The anti-PD-1 agent nivolumab also received FDA approval after achieving similar results in a phase 3 study at a dose of 3 mg/kg once every 2 weeks compared with chemotherapy (Robert et al., 2015).

The attempt to synergize the efficacy of anti-CTLA-4 and anti-PD-1 was realized with the FDA approval in 2015 of the ipilimumab/nivolumab combination. The regimen being investigated in the CheckMate 067 trial, by using nivolumab 1 mg/kg and ipilimumab 3 mg/kg once every 3 weeks for four cycles followed by maintenance nivolumab 3 mg/kg once every 2 weeks. Patients were randomly assigned in a 1:1:1 ratio to receive the combination treatment, nivolumab 3 mg/kg (plus ipilimumab-matched placebo), or ipilimumab 3 mg/kg (plus nivolumab-matched placebo). Co-primary endpoints were OS and PFS. Moreover, PFS has been compared between the nivolumab group or the ipilimumab/nivolumab combination group versus the ipilimumab single agent group. The study was not designed for a formal statistical comparison between the nivolumab and the ipilimumab/nivolumab groups. This trial randomized 945 patients and resulted in unprecedented survival results, reaching a mOS of more than 6 years, becoming the main regimen of choice in advanced melanoma irrespective of BRAF mutational status (Hodi et al., 2022). Moreover, the ORR was 60% in the combination group (with a 22% complete response rate), 45% in the nivolumab group and 20% in the ipilimumab group. Median PFS was 11.5 months in the combination group, 7 months in the nivolumab group, and 3 months in the ipilimumab group (Larkin et al., 2019). However, this benefit comes at the cost of a grade 3–4 irAEs rate of around 60% (Larkin et al., 2015). In the attempt to lower the incidence of irAEs while maintaining the same efficacy, clinical trials have been conducted to adjust the dosage of the drugs in the combination (Lebbé et al., 2019). One study observed that switching doses to nivolumab 3 mg/kg and ipilimumab 1 mg/kg for four doses followed by nivolumab maintenance could lower the incidence of grade 3–4 irAEs by 14%, with overlapping PFS and OS curves (Lebbé et al., 2019). Another study tested standard-dose pembrolizumab associated with alternative-doses and schedules ipilimumab (1 mg/kg every 3 weeks for four cycles; 50 mg every 6 weeks for four doses; and 100 mg every 12 weeks for four doses). The results showed severe treatment-related AEs (grade 3–5) rate of 47%, 24%, and 39%, respectively, with similar ORRs to standard-dose ipilimumab/nivolumab (Carlino et al., 2020; Long et al., 2021). However, these regimens have not entered clinical practice due to a lack of enough mature follow-up, and the fear of not having as durable responses as the classic schedule.

Recently, a new combination of ICIs, relatlimab/nivolumab was approved by the FDA. Relatlimab is a first-in-class inhibitor of Lymphocyte-activation gene 3 (LAG-3), a surface molecule inhibiting T-lymphocyte proliferation (Tawbi et al., 2022). The RELATIVITY-047 study demonstrated the superiority of relatlimab/nivolumab over nivolumab monotherapy, doubling the PFS (approximately 10 months versus 5 months), but most importantly showing a grade 3–4 irAEs profile of less than 20% (Tawbi et al., 2022). These results, albeit still immature, made the choice of first-line therapy in metastatic melanoma more challenging. This choice has to be based both on clinical features, such as disease burden, presence of melanoma-associated symptoms, brain metastases, performance status and pre-existing medical issues of the patient and on the basis of molecular features such as BRAF mutation, PD-L1 expression and laboratory characteristics (such as LDH levels) (Boutros et al., 2023). Finally, it should not be ignored that today many patients may have already received ICIs in the neoadjuvant or adjuvant setting before being diagnosed with metastatic disease, and this could certainly undermine the choice of treatment (Switzer et al., 2023).

Given the long follow-up, studies are underway to assess the hypothesis of maintenance of clinical benefit upon discontinuation of therapy. This hypothesis arose from the discontinuation of treatment due to toxicity. A post-hoc analysis estimated PFS and OS in patients treated with combination therapy who discontinued treatment (ipilimumab/nivolumab) because of treatment-related adverse event during

the induction phase (Hodi et al., 2018). Differences between the two groups were not statistically significant for either PFS or OS (Hodi et al., 2018). Interestingly, median time to subsequent therapy was not reached for ipilimumab/nivolumab, it was 24.7 months with nivolumab, and 8 months with ipilimumab (Wolchok et al., 2022). The median treatment-free interval (the time from the last dose of study drug to initiation of subsequent systemic therapy or the last known date alive) was 27.6, 2.3, and 1.9 months in the three groups, respectively (Wolchok et al., 2022).

The post-hoc 5-year analysis of the KEYNOTE-006 trial investigated the PFS from cessation of pembrolizumab in patients who completed 2 years of treatment (Robert et al., 2019). The best results were obtained in patients who achieved complete or partial responses (with about 85% of 2-years PFS) compared to patients with stable disease (where the 2-years PFS is less than 50%) (Robert et al., 2019).

A recent European Society of Medical Oncology (ESMO) consensus stated that patients who were confirmed to be in complete response may be considered for stopping treatment after at least 6 months of anti-PD-1 therapy (Keilholz et al., 2020). In case of partial response and stable disease patients may be considered for stopping treatment after 2 years of treatment (Keilholz et al., 2020). According to the 2023 National Comprehensive Cancer Network (NCCN) Guidelines, despite the optimal duration of anti-PD-1 therapy remains unknown, most patients who achieve an objective response and stop anti-PD-1 after 2 years of therapy maintain the response within 2 years of follow-up (NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) Melanoma: Cutaneous Version 2. 2023 — March 10, 2023 n.d.).

4. BRAF-mutant melanoma: approved combinations and treatment sequencing

About half of all patients with metastatic melanoma have a BRAF-mutant melanoma. BRAF is a gene involved together with MEK in the mitogen-activated protein kinase (MAPK) signal transduction pathway (Flaherty et al., 2010). Alteration of this signal leads to uncontrolled cell proliferation. In particular, the most common BRAF mutation in melanoma occurs with the substitution of a valine (V) for a glutamic acid (E) in the 600 position (Flaherty et al., 2010).

Vemurafenib and dabrafenib are the first inhibitors targeting BRAFV600, and demonstrated in two studies a response rate of about 50% (Chapman et al., 2011; Hauschild et al., 2012). However, these results were not lasting due to acquired resistance mechanisms leading to reactivation of the MAP kinase pathway (Switzer et al., 2022). Thus, with the combination of BRAF and MEK inhibitors the onset of these resistances has been delayed (as well as having led to a reduction of the skin toxicity which was associated with BRAF inhibitors in monotherapy). The BRAF/MEK inhibitor combinations dabrafenib/trametinib and vemurafenib/cobimetinib achieved median PFS of about one year (Larkin et al., 2014; Long et al., 2015) and an ORR as high as 70%, with potential long-term benefit reaching a 5-years OS of about 35% (Long et al., 2017; Ascierto et al., 2021). Similar results were obtained with the encorafenib/binimetinib combination (Dummer et al., 2018), with manageable toxicity profiles. While some toxicities are common to all BRAF/MEK inhibitors combinations (such as nausea, fatigue, and the ocular toxicities associated with MEK inhibitors), photosensitivity and diarrhea are frequent with vemurafenib/cobimetinib but rare with dabrafenib/trametinib, while fever and chills are mostly associated with dabrafenib/trametinib (Larkin et al., 2014; Long et al., 2015; Dummer et al., 2018).

The main advantage of these combinations is the achievement of rapid responses with significant tumor shrinkage in most patients. This feature is clinically relevant especially for patients with severe melanoma-associated symptoms (Tanda et al., 2020).

In other cases, all possible treatment options remain open: frontline targeted therapy or ICIs. The DREAMSeq and SECOMBIT studies were conducted to determine the best sequence of these treatments. In the

phase 3 DREAMSeq trial, 265 BRAF-mutant melanoma patients were randomized to receive either first-line ipilimumab/nivolumab followed by dabrafenib/trametinib at disease progression, or the reverse sequence (Atkins et al., 2022). The study established that the sequence beginning with combination ipilimumab/nivolumab resulted in a 20% absolute improvement in 2-year overall survival (72% versus 52%) as compared with first-line dabrafenib/trametinib followed by immunotherapy (Atkins et al., 2022). This finding is supported by the superiority in duration of response with the immunotherapy-first sequence strategy (88% of the responses ongoing compared with less than 50% with dabrafenib/trametinib) (Atkins et al., 2022). On the other hand, the comparable clinical activity of targeted therapy in first and in second line (in contrast to immunotherapy), encourages its use in the second-line setting. In fact, immunotherapy showed an absolute 15% decrease in terms of activity in second-line, with an ORR of 30% (Switzer et al., 2023). This study had some limitations, mainly due to the open-label design and slow accrual (265 patients over 6 years), and the crossover rate of only 52% (Atkins et al., 2022).

The phase 2 SECOMBIT study randomized around 200 BRAF-mutant patients into three treatment arms corresponding to three different sequencing strategies. In arm A, patients received first-line encorafenib/binimetinib until disease progression, followed by ipilimumab/nivolumab; in arm B they received first-line ipilimumab/nivolumab and - at disease progression - encorafenib/binimetinib; while in arm C patients received short course (8 weeks) encorafenib/binimetinib, followed by ipilimumab/nivolumab, and - at disease progression - encorafenib/binimetinib (Ascierto et al., 2022). The primary endpoint was OS (Ascierto et al., 2022). Despite the study was not statistically powered to formally compare the three arms of treatment, the results showed better survival results in patients randomized in arms B and C compared to arm A, with an increase in 4-year survival of more than 15% between arm B and arm A (Ascierto et al., 2022). Furthermore, as described in DREAMSeq trial, response rates to targeted therapy were not significantly affected from the first to second line of treatment, in contrast to that observed with immunotherapy (Ascierto et al., 2022). Moreover, the strategy including a short course of targeted therapy before immunotherapy (arm C), showed a 10% superiority in both 4-year PFS and OS compared to arm B in the subgroup of patients with high levels of LDH, suggesting that this regimen could be worth further investigation especially in patients with bad prognostic features (Ascierto et al., 2022). The main limitations of this study are the lack of formal testing between arms and the open-label design (Ascierto et al., 2022).

Despite their limitations, these are the first studies investigating the best treatment strategy for advanced BRAF-mutant melanoma, with OS as the primary endpoint (Atkins et al., 2022; Ascierto et al., 2022). Notably, both studies obtained similar 2-years OS results and similar trends towards a better long-term OS and PFS with the first-line combination of ICIs (Atkins et al., 2022; Ascierto et al., 2022), and these results were confirmed across all described subgroups (age, gender, performance status, LDH level or disease stage) (Atkins et al., 2022; Ascierto et al., 2022).

In the attempt to synergize the effect of BRAF/MEK inhibitors and anti-PD-(L)1 to combine the rapidity of response of targeted therapies with the duration of response of immunotherapy, three combination triplets were tested (Gutzmer et al., 2020; Ferrucci et al., 2020; Dummer et al., 2022). However, the only one to receive FDA approval was atezolizumab combined with vemurafenib/cobimetinib, showing in the phase 3 study IMspire150 a statistically significant improvement of about 5 months in median PFS compared to targeted therapy alone (15 versus 10 months, HR=0.78 95%CI 0.63 - 0.97) (Gutzmer et al., 2020). However, the clinical significance of this benefit was not clear, as patients in the experimental arm had a more than two times increased risk of treatment-related toxicities compared to control (Gutzmer et al., 2020). The other two triplets investigated (pembrolizumab/dabrafenib/trametinib and spartalizumab/dabrafenib/trametinib) showed a clinical benefit in terms of both PFS and duration

of responses compared to targeted therapy alone, although they did not meet their primary endpoints (Ferrucci et al., 2020; Dummer et al., 2022). It should be noted that PFS curves tended to separate in favor of triplet arms after 6–8 months, and this may have affected their success in reaching the primary endpoint at the time of the analysis (Gutzmer et al., 2020; Ferrucci et al., 2020; Dummer et al., 2022). On the other hand, one of the limitations of these three studies is the absence of a comparator arm with immunotherapy (Gutzmer et al., 2020; Ferrucci et al., 2020; Dummer et al., 2022).

To overcome this limitation, the ongoing STARBOARD study (NCT04657991) is evaluating the efficacy of pembrolizumab/encorafenib/binimetinib compared to pembrolizumab alone. Therefore, the role of combination triplets in metastatic melanoma still needs to be defined.

5. Management of brain metastases

Brain metastases frequently occur in patients with melanoma, leading to a deterioration of the prognosis. Melanoma ranks third among the most prevalent neoplasms responsible for causing brain metastases (Berghoff et al., 2016). In certain case series, brain metastases have been observed in melanoma patients with a frequency as high as 30%, resulting in a survival prognosis of approximately 6 months (Cagney et al., 2017).

The integration of new drug classes, such as immunotherapy and targeted therapy for patients with BRAF mutations, has significantly revolutionized the management and prognosis of this specific patient population (Spagnolo et al., 2016). While there are no direct comparisons between targeted therapy and immunotherapy in these patients, the current recommendations are based on indirect analyses of trials assessing each class. The preferred initial treatment for patients with melanoma and asymptomatic brain metastases is the combination of the immunotherapeutic agents ipilimumab (3 mg/kg) and nivolumab (1 mg/kg) for four cycles, followed by maintenance treatment with nivolumab, regardless the BRAF status. The activity of this drug combination as a first-line approach has been investigated in three clinical studies.

In the phase 2 CheckMate 204 trial, two cohorts of patients received the ipilimumab/nivolumab combination. Asymptomatic patients were included in cohort A (n = 101) and symptomatic patients in cohort B (n = 18). Symptomatic patients, with stable neurological symptoms, were allowed to receive low-dose dexamethasone (≤ 4 mg) at baseline (Tawbi et al., 2021). In cohort A, the primary endpoint of intracranial clinical benefit (including complete responses, partial responses, and stable disease lasting ≥ 6 months), was achieved in 58 out of 101 patients (57.4%). However, in cohort B, only four out of 18 patients (22.2%) demonstrated the same intracranial clinical benefit. Two out of the four patients, who had an intracranial tumor burden measuring less than 1 cm, attained a complete intracranial response. The remaining two patients, with an intracranial tumor burden of approximately 2 cm, demonstrated an intracranial partial response. Furthermore, among the four patients showing positive responses, three exhibited a performance status of 1, and only two were under baseline corticosteroid treatment (at a dosage not exceeding 2 mg of dexamethasone). The patients who responded intracranially also responded extracranially. The median intracranial PFS for cohort A was 39.3 months, whereas cohort B had a significantly lower median intracranial PFS of 1.2 months (Tawbi et al., 2021); however, all the four patients with an intracranial response had durable and ongoing responses with a minimum follow-up of 6 months (Tawbi et al., 2021).

In the ABC trial, an open-label phase 2 study, the activity of nivolumab as monotherapy or in combination with ipilimumab was assessed in both symptomatic (n = 16) and asymptomatic (n = 60) patients with melanoma brain metastases. Asymptomatic patients were randomly assigned to either cohort A (n = 35), receiving ipilimumab/nivolumab or cohort B (n = 25), receiving nivolumab monotherapy. Symptomatic patients (n = 16), with failed local therapy, neurological symptoms, or

leptomeningeal disease, were enrolled in the non-randomized cohort C, receiving nivolumab monotherapy (Long et al., 2018). The results further support the use of combination immunotherapy in patients with asymptomatic disease. According to the last updated 5-year follow-up results, out of 76 patients, intracranial response (ICR) rates were 51% in cohort A, 20% in cohort B, and 6% in cohort C. The 5-year intracranial PFS rates were 46% in cohort A, 15% in cohort B, and 6% in cohort C (Long et al., 2021).

In the phase 3 NIBIT M2 study, a total of 23 patients received fotemustine, 26 patients were treated with ipilimumab plus fotemustine, and 27 patients with the ipilimumab/nivolumab combination. The observed intracranial response rates were 0%, 19% and 44% in the three treatment arms, respectively. The median OS was 29.2 months in the ipilimumab plus nivolumab arm. Moreover, the ipilimumab plus nivolumab arm showed a significantly higher four-year survival rate (41.0%) compared to fotemustine (10.9%). Treatment-related adverse events were observed in 48% of patients in the fotemustine arm, 69% in the ipilimumab plus fotemustine arm, and 30% in the ipilimumab plus nivolumab arm (Di Giacomo et al., 2021).

The consistent results of these studies highlight the high efficacy of ipilimumab/nivolumab in patients with asymptomatic brain metastases. However, the poor outcomes observed in cohorts involving symptomatic patients, although based on a small number of subjects, emphasize the significant unmet medical need in treating symptomatic brain metastases.

In patients with BRAF mutations, BRAF/MEK inhibitors were also tested. In the phase 2 COMBI-MB trial, a total of 125 patients with brain metastases were treated with dabrafenib/trametinib. Out of these, 17 patients had symptomatic brain metastases. The primary endpoint was the investigator assessed intracranial response. A response rate of approximately 60% was observed in all patients, including those with symptomatic brain metastases. However, the median PFS was around 5 months for all patients (except for those with BRAFV600E mutation who underwent prior local treatments, where it was approximately 7 months) (Davies et al., 2017).

In light of these findings, the 2020 ESMO consensus conference suggested a treatment approach for BRAF-mutant melanoma brain metastases: immunotherapy combination was recommended for asymptomatic patients, while targeted therapy was advised for those with symptoms (Keilholz et al., 2020).

In an effort to further enhance these results, the strategy of combination of targeted therapy with ICIs was also tested. The TRICOTEL study aimed to evaluate the combination of atezolizumab with cobimetinib (in a BRAF wild-type cohort, early closed, and in which the primary endpoint was not analyzed) or vemurafenib/cobimetinib (in a BRAF-mutant cohort) in patients with melanoma and central nervous system metastases. The primary endpoint was the independent committee reviewed intracranial objective response rate. The response had to be confirmed by a 4-weeks apart assessment. Results showed that the intracranial objective response rate was 42% in the BRAFV600 mutation-positive cohort. The median intracranial PFS was about 7 months. The unique aspect of this trial was the possibility to include patients with symptomatic brain metastases (Dummer et al., 2023).

In a post-hoc analysis of patients with baseline symptoms or receiving corticosteroids ($n = 26$), the intracranial objective response rate was 35% according to independent review committee assessment and 38% according to investigator assessment. The median duration of intracranial response was around 7.4 months (Dummer et al., 2023).

LDH: lactate dehydrogenase; ULN: upper normal limit; wt: wild-type. Created with BioRender.com.

6. Non-cutaneous melanoma

These entities represent distinct clinical conditions from cutaneous melanoma, characterized by distinct epidemiology, molecular patterns, including tumor mutational burden, PD-L1 expression, and the presence

of BRAF mutations, as well as different clinical and prognostic characteristics in terms of response to standard treatments (McLaughlin et al., 2005).

6.1. Uveal melanoma

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults, accounting for approximately 3–5% of all melanomas. It is considered a rare neoplasm, with an incidence ranging from 0 to 6 cases per million individuals (Singh et al., 2011; Hu et al., 2005). UM originates from melanocytes and shares certain risk factors with cutaneous melanoma, including fair skin and light eyes. However, it is also associated with ocular melanosis, the presence of atypical cutaneous nevi, common cutaneous nevi, iris nevi, and specific genetic mutations. These factors contribute to the development and progression of UM (Shields et al., 2013; Gallagher et al., 1985). Localized UMs can often be effectively treated with surgical intervention or radiotherapy. However, it is important to note that about half of the patients may eventually experience distant metastases. Among these metastases, the liver is the most frequently affected site (Diener-West et al., 2005).

Despite advancements in the treatment of cutaneous melanoma, the application of these treatments (in particular ICIs) to metastatic ocular melanoma has yielded limited clinical benefits. As a result, the prognosis for ocular melanoma remained consistently poor, with a median survival of approximately one year from the time of metastatic disease diagnosis (Kuk et al., 2016). A retrospective study conducted on 56 patients treated with anti-PD-1 or anti-PD-L1 monotherapy between 2009 and 2015 revealed relatively poor outcomes. Only 3.6% of the patients achieved an objective response, with a 9% of patients achieving disease stability. The median PFS in this study was 2.6 months, suggesting limited efficacy of monotherapy with anti-PD-1 or anti-PD-L1 agents in this particular cohort of patients (Algazi et al., 2016). In the Italian Expanded Access Program, 82 pre-treated patients had been treated with ipilimumab, showing an ORR of 5%, with median PFS and OS of 3.6 and 6.0 months, respectively (Maio et al., 2013). The ipilimumab/nivolumab in combination was tested in the phase 2, single arm, GEM-1402 trial. A total of 52 patients received the combination of anti-PD-1 and anti CTLA-4 with an ORR of 11.5% of patients. The most common outcome was stable disease (52%), while median PFS and OS were 3 months and 12.7 months, respectively (Piulats et al., 2021).

In a recent phase 3 multicenter open-label study, tebentafusp, a bispecific protein composed of a T lymphocyte receptor fused with an anti-CD3 effector, has shown promising effectiveness in patients with metastatic UM. The study enrolled 378 patients who tested positive for HLA-A* 02:01 and had not been previously treated for metastatic disease. These patients were randomized to receive either tebentafusp or the clinician's therapeutic choice (Nathan et al., 2021).

The results of the study demonstrated that patients treated with tebentafusp achieved a significant improvement in median overall survival compared to the control group. The median overall survival was 21.7 months in the tebentafusp group, compared to 16 months in the control group (HR 0.51, 95% CI 0.37 - 0.71). Furthermore, at a follow-up of 6 months, tebentafusp showed a progression-free survival rate of 31%, whereas the control treatment had a rate of 19% (HR 0.73, 95% CI 0.58 - 0.94). The study also reported an objective response rate of 9% in the tebentafusp group, compared to 5% in the control group (Nathan et al., 2021).

It's important to note that grade 3/4 reactions were observed in 44% of patients treated with tebentafusp. These reactions were mainly related to drug-induced cytokine release, such as pyrexia in 4% of patients, hypotension in 3%, and cytokine release syndrome in 1%. Skin reactions, including rash in 18% of patients and pruritus in 4%, were also documented (Nathan et al., 2021).

These findings suggest that tebentafusp holds promise as a novel treatment option for patients with metastatic UM. Despite the occurrence of grade 3/4 reactions, the improvement in OS and PFS rates,

along with objective responses, indicate the potential benefit of tebentafusp in this patient population. Further research and long-term follow-up are needed to better understand the safety profile and long-term efficacy of tebentafusp in the treatment of metastatic UM (Nathan et al., 2021).

Liver is the most frequent site of metastasis from UM, therefore locoregional approaches have been studied for this category of patients in the so called 'liver directed therapy'.

The randomized phase 3 FOCUS trial studied the percutaneous hepatic perfusion (PHP) with melpahalan in patients with hepatic-dominant melanoma. Treatment naïve patients were randomized to receive PHP (n = 102) or clinician's choice therapy (n = 42). Preliminary results showed an ORR of 35% in the PHP group and 12.5% in the control group, with a disease control rate of 73.6% in the PHP group versus 37.5% in the control group. Median PFS was 9 versus 3 months in the PHP and in the control group, respectively. Median OS was 20 and 14 months in the PHP and in the control group, respectively (Zager et al., 2022).

Similar data were obtained in the SCANDIUM phase 3 trial, where isolated hepatic perfusion (IHP) with melphalan showed superior overall response rates (40% versus 4.5%) and prolonged PFS (7.4 months vs. 3.3 months) compared to best alternative care in patients with previously untreated isolated liver metastases from UM. The median hepatic PFS was also significantly longer in the IHP group (9.1 months vs. 3.3 months). Adverse events were generally manageable, with a low incidence of treatment-related serious adverse events. These findings highlight the efficacy and safety of IHP as a regional treatment option for this patient population (Olofsson Bagge et al., 2023).

These data suggest that, in the presence of metastatic UM, along with the search for HLA-A* 02:01 for tebentafusp, if feasible, the liver-directed therapy approach should be considered.

6.2. Mucosal melanoma

Mucosal melanoma (MM) is a rare form of melanoma, comprising less than 2% of all melanomas. It primarily occurs in mucous membranes, with the most common sites being the oral cavity, nasal passages, anorectal area, genital mucosa, and the conjunctiva. Unlike cutaneous melanoma, MM is not typically associated with the presence of melanocytic nevi. MM tends to affect women more frequently than men, although the reasons for this gender disparity are not fully understood. Due to its rarity and unique characteristics, MM poses challenges in terms of diagnosis, treatment, and prognosis (Hicks and Flaitz, 2000; Chang et al., 1998).

Surgery is the main treatment for localized mucosal melanoma, with possible lymphadenectomy for lymph node metastases. Adjuvant radiotherapy can help control local recurrences, but there is limited evidence of clinical benefit. The use of adjuvant therapy in mucosal melanoma remains uncertain, and decisions should be based on individual factors. Further research is needed to determine its efficacy (Krengli et al., 2006; Temam et al., 2005; Lian et al., 2013).

Studies have suggested that metastatic (MM) may show less benefit from immunotherapy treatments compared to metastatic cutaneous melanoma. MM generally exhibits a lower tumor mutational burden (TMB) compared to cutaneous melanoma. This is attributed to the fact that MM is not primarily driven by DNA mutations caused by chronic ultraviolet sun exposure, which is a major factor in CM. These molecular differences between the two subtypes may contribute to variations in their response to standard treatments.

A pooled analysis of clinical studies included 889 patients receiving nivolumab monotherapy. Among them, 86 (10%) had mucosal melanoma and 665 (75%) had cutaneous melanoma. The median PFS was 3 months for MM and 6 months for cutaneous melanoma, with objective response rates of 23.3% and 40.9%, respectively. In patients receiving nivolumab combined with ipilimumab, the median PFS was 5.9 months for mucosal melanoma and 11.7 months for cutaneous melanoma, with

objective response rates of 37.1% and 60.4%, respectively (D'Angelo et al., 2017). Moreover, in the RELATIVITY-047 trial, the ICIs combination of anti-PD-1 and anti-LAG-3 did not show a statistically significant benefit compared to anti-PD-1 alone. In this study, a total of 51 patients with metastatic MM received relatlimab/nivolumab (n = 23) or nivolumab (n = 28), with a trend for a benefit in PFS (HR 0.72, 95% CI 0.36 - 1.45) (Tawbi et al., 2022).

MMs exhibit a distinct genomic profile compared to cutaneous melanomas. In particular, the frequency of c-KIT mutation ranges about 20% in mucosal melanomas and 15% in acral melanomas (Curtin et al., 2006). Assessment of c-KIT mutational status is recommended in acral and mucosal melanomas, following evaluation of BRAF mutational status (since both mutations are mutually exclusive). The most commonly associated c-KIT mutations in melanoma include L576P in exon 11 and K642E in exon 13, with other mutations described such as V599A, D816H, and D820Y. Limited phase 2 clinical experiences have shown objective responses in melanomas harboring c-KIT mutations in exons 9, 11, or 13 when treated with c-KIT inhibitors. A phase 2 study by Kim in 2008 evaluated imatinib treatment in 21 patients with advanced melanoma and mutation in a Protein Tyrosine Kinase (c-kit, PDGFR, c-abl, or abl-related gene), resulting in a single objective response in a c-KIT mutant patient (Kim et al., 2008). Another phase 2 study by Hodi in 2013 examined 24 patients with advanced melanoma and c-KIT mutation or amplification treated with imatinib, achieving a response rate of 29% and a median PFS of 3.7 months (Hodi et al., 2013). Carvajal et al.'s phase 2 study assessed 28 patients with advanced melanoma and c-KIT mutations treated with imatinib, yielding a response rate of 16% (in patients harboring mutations in exon 9 or 11) (Carvajal et al., 2011). These studies are limited by small sample sizes, and further research is necessary to determine the optimal treatment approach for these specific melanoma subtypes.

The efficacy of alternative inhibitors and their activity on brain metastases in melanomas after progression to imatinib is unknown. A phase 2 study investigated nilotinib in two cohorts of patients with KIT mutations: those refractory or intolerant to a prior KIT inhibitor (cohort A) and those with brain metastases (cohort B). Disease control was achieved in a subset of patients, but the efficacy of nilotinib in KIT-altered melanoma with brain metastasis is limited. Out of the 19 treated patients (11 in cohort A, 8 in cohort B), partial responses were observed in two patients in cohort A (18.2%). No responses were observed in cohort B (Carvajal et al., 2015).

In a retrospective study involving 34 patients with metastatic conjunctival melanoma, frequent genetic alterations were observed, including BRAF mutations in 46.7% of cases. Other mutations identified were NRAS, NF1, and TERT promoter mutations. Patients treated with first-line BRAF/MEK inhibitors had a one-year PFS rate of 54.7% and median OS of 29.1 months (Lodde et al., 2022).

Recent studies are suggesting a role for antiangiogenic drugs in combination with ICIs for the treatment of advanced MM.

In a phase 1b trial, the combination of toripalimab and axitinib was assessed in 33 chemotherapy-naïve patients with metastatic MM. The combination treatment demonstrated an ORR of 48.3%, with a median OS of 20.7 months and a median PFS of 7.5 months (Li et al., 2022). In line with these findings, the combination of atezolizumab plus bevacizumab has demonstrated activity in advanced MM in a phase 2 study conducted in 2022. A total of 40 previously untreated patients were evaluated with an ORR of 45% and a median PFS of 8.2 months (Mao et al., 2022). Similar results were obtained with the combination of cambrelizumab plus apatinib. In this study, 30 patients were treated with an ORR of 43% and a median PFS of 7.2 months (Zou et al., 2022). The observed objective responses and median progression-free survival are promising, suggesting that these combination therapies could provide significant benefit to patients with advanced mucosal melanoma. Further studies and research are needed to confirm and further investigate these promising results.

6.3. Desmoplastic melanoma

Desmoplastic melanoma is a distinct subtype of cutaneous melanoma that exhibits unique clinical and histological features. It typically occurs in sun-exposed areas and is associated with chronic UV exposure. Unlike other melanomas, desmoplastic melanoma has a higher propensity for local recurrence and a lower incidence of lymph node metastases. It presents challenges in diagnosis due to its amelanotic nature and requires careful histopathological examination. The tumor is characterized by spindle cells and abundant collagen, and it displays a deeply infiltrative growth pattern with a notable neurotropism (Chen et al., 2013). A retrospective study conducted in 2018 provided evidence that immunotherapy may be highly effective in the treatment of desmoplastic melanoma. The study included 60 patients with advanced desmoplastic melanoma who were treated with PD-1 or PD-L1 inhibitors. The results showed a significant proportion of responders (70%), with a 32% complete response rate (Eroglu et al., 2018). These findings highlight the effectiveness of ICIs in treating desmoplastic melanoma. The efficacy of ICIs was further supported in a prospective phase 2 study, where 27 patients with resectable desmoplastic melanoma received pre-operative pembrolizumab. The study reported a 56% pathological complete response rate, confirming the previous findings (Kendra et al., 2022).

LDH: lactate dehydrogenase; ULN: upper normal level.
LDH: lactate dehydrogenase; ULN: upper normal level.

7. Treatment algorithm for advanced melanoma

With the expansion of therapeutic indications into the advanced (Table 2 (Robert et al., 2015; Larkin et al., 2019; Larkin et al., 2015; Tawbi et al., 2022; Hodi et al., 2018; Wolchok et al., 2022; Robert et al., 2019; Larkin et al., 2014; Long et al., 2015; Long et al., 2017; Ascierto et al., 2021; Gutzmer et al., 2020; Dummer et al., 2022; Wolchok et al.,

2017; Long et al., 2022; McArthur et al., 2022; Robert et al., 2020; Ascierto et al., 2019; Robert et al., 2014; Schachter et al., 2017; Dummer et al., 2022; Ascierto et al., 2020; Robert et al., 2019; Long et al., 2014; Grob et al., 2015; Robert et al., 2015; Ascierto et al., 2016)) and adjuvant setting, and in light of recent studies on sequencing and combination strategies, the potential clinical scenarios that may arise in everyday clinical practice have significantly increased. In Figs. 1- 3 we have presented some of the most common clinical scenarios and a potential treatment algorithm in treatment naïve patients (or those who had late recurrence after more than six months after treatment completion) and in patients who relapsed during adjuvant treatment or within six months after the completion of adjuvant therapy (Figs. 2–3).

In fact, with the establishment of adjuvant therapy in clinical practice, this issue has become even more significant, as there are currently limited therapeutic salvage strategies available for these patients. Despite the observed prolongation of recurrence-free survival with anti-PD-1 antibodies, approximately 25%– 30% of patients experience recurrence within one year. A recent study investigated the pattern of recurrence, management strategies, and outcomes in patients who recurred despite receiving adjuvant anti-PD-1 therapy. Among approximately 850 patients treated, melanoma recurrence occurred in 17% (147 patients). Most recurrences occurred during adjuvant anti-PD-1 treatment (76%), with a smaller proportion observed after treatment cessation (24%). Treatment responses varied depending on the timing of recurrence and the therapeutic regimen employed. Importantly, the findings suggest limited activity of further anti-PD-1 monotherapy in patients who experienced recurrence during PD-1 treatment (no responses observed), highlighting the potential clinical utility of alternative strategies such as ipilimumab (24% responded to ipilimumab alone or in combination with anti-PD-1) and BRAF/MEK inhibitors (observed ORR=78%). Retreatment with anti-PD-1 may hold potential benefit for select patients who experienced recurrence after discontinuing anti-PD-1 therapy (observed ORR=40%) (Owen et al., 2020).

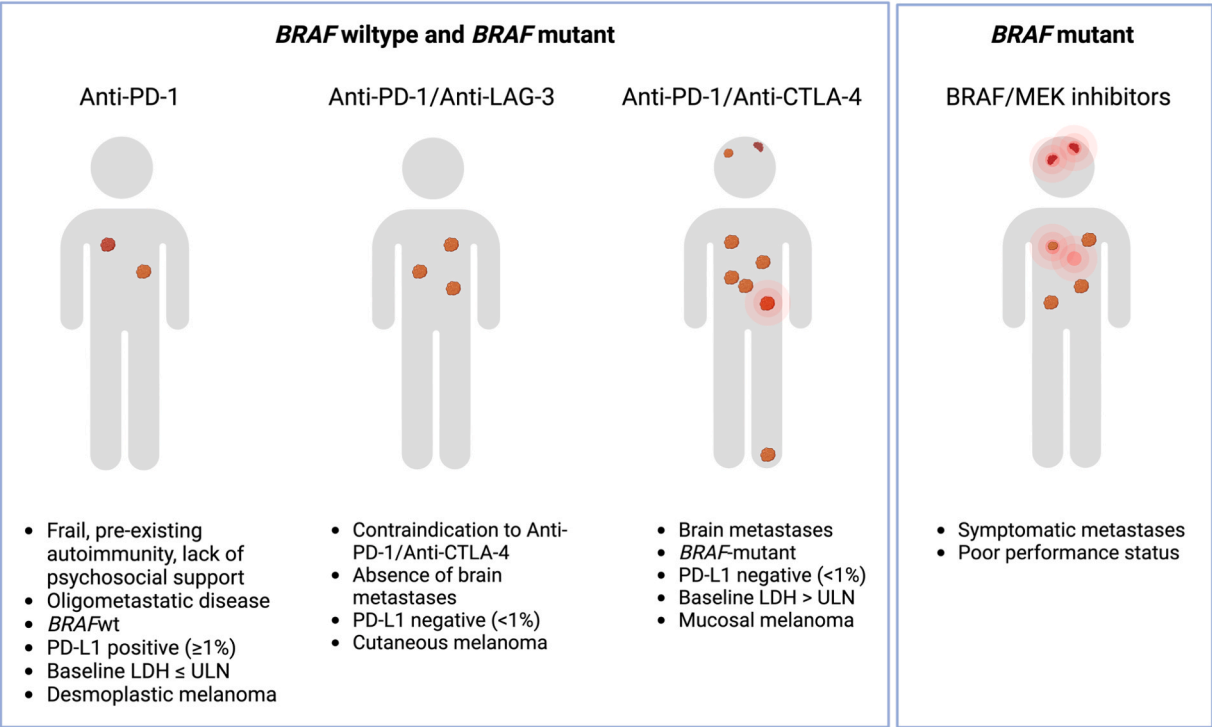


Fig. 1. Different clinical scenarios of metastatic melanoma, guiding therapeutic choices based on clinical, histological, and molecular characteristics. The figure depicts four human silhouettes, each representing different advanced melanoma clinical scenario, as indicated by the red spots. The first silhouette shows a couple of spots, suggesting a low-volume disease. The second silhouette has a few more spots, indicating a higher number of metastatic sites. The third silhouette shows an increased number of metastatic sites, including the brain and symptomatic lesions (marked by the red halo). The fourth silhouette shows a critical burden of disease with extensive and symptomatic lesions, also involving the brain.

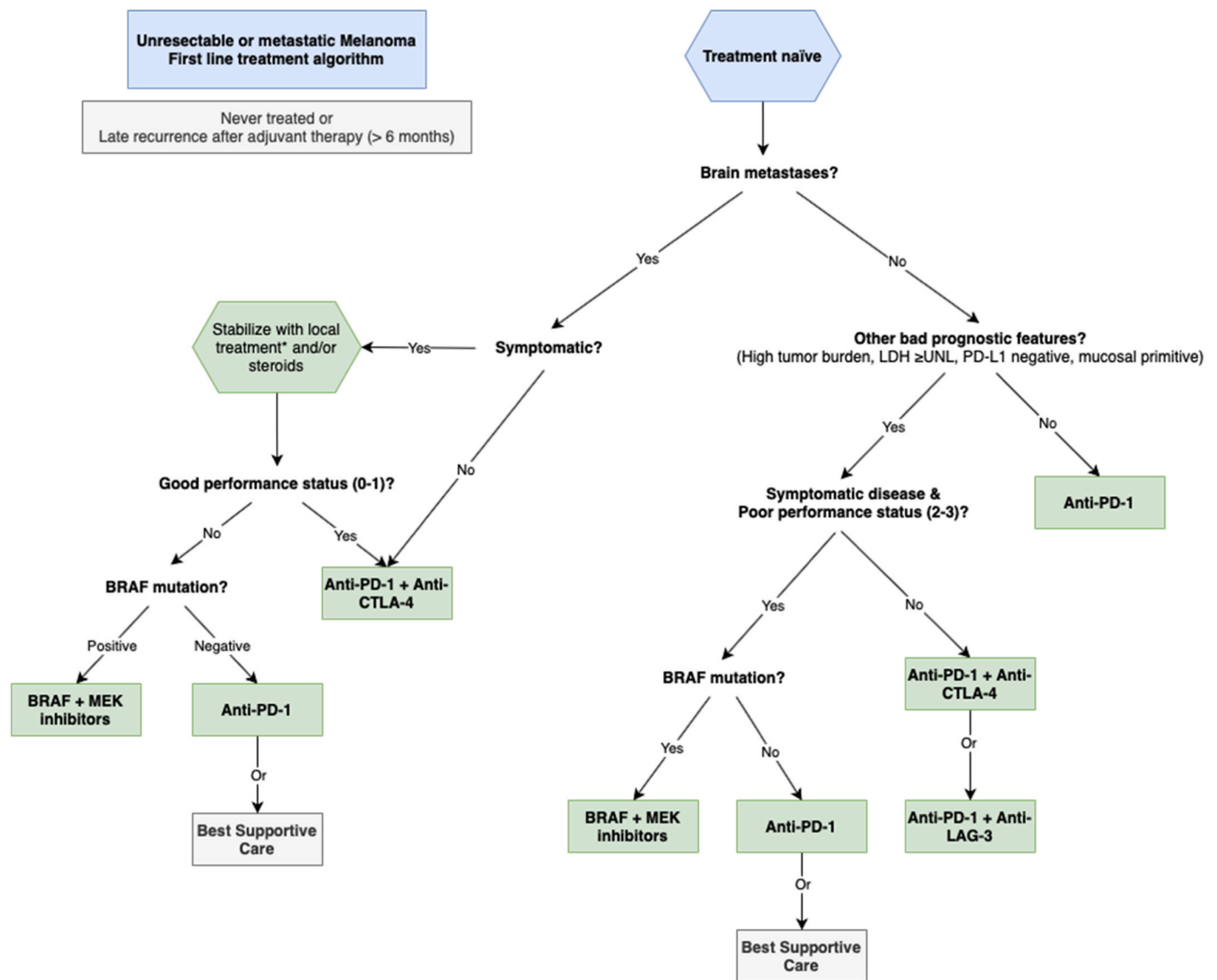


Fig. 2. Advanced Melanoma first-line treatment algorithm for treatment naïve patients, including those who had late recurrence after adjuvant therapy (i.e. after more than six months from the completion of adjuvant treatment). In this subgroup of patients the main decision point depends on the presence of brain metastases and other bad prognostic features.

To face this clinical issue, a retrospective cohort study investigated the efficacy and safety of ipilimumab, either as monotherapy or in combination with anti-PD-1 (pembrolizumab or nivolumab), in patients with metastatic melanoma who were resistant to anti-PD-(L)1 therapy. The study included 355 patients and found that the combination of ipilimumab plus anti-PD-1 showed higher efficacy compared to ipilimumab monotherapy. The ORR was significantly higher with ipilimumab plus anti-PD-1 (31% of 193 patients) than with ipilimumab monotherapy (13% of 162 patients). Overall survival was longer in the combination group (mOS 20.4 months) than with monotherapy (mOS 8.8 months). The rate of grade 3–5 adverse events was similar between the two groups (approximately 30%). These findings support the use of ipilimumab plus anti-PD-1 as a preferred second-line immunotherapy for patients with advanced melanoma resistant to anti-PD-(L)1 therapy (Pires da Silva et al., 2021).

Furthermore, a prospective clinical trial aimed to evaluate the combination of low dose ipilimumab (1 mg/kg) and pembrolizumab as a treatment option for patients with advanced melanoma who had experienced progression on prior anti-PD-(L)1 immunotherapy. The study enrolled a total of 70 patients, and the primary endpoint was ORR. The

results showed that the combination therapy achieved an overall response rate of 29% in the entire trial population, with a median progression-free survival of 5.0 months and a median overall survival of 24.7 months. The median duration of response was 16.6 months. The combination treatment demonstrated significant antitumor activity and tolerability, including responses in patients with PD-L1-negative tumors and non-T-cell-inflamed phenotypes. This study provides evidence for the effectiveness of pembrolizumab plus low-dose ipilimumab after progression on anti-PD-(L)1 immunotherapy in advanced melanoma patients (Olson et al., 2021). Another phase 2 prospective trial randomized 91 patients with metastatic melanoma whose disease had progressed on front-line anti-PD-(L)1 therapy, to receive in a 3:1 ratio the combination ipilimumab/nivolumab or ipilimumab alone (VanderWalde et al., 2023). The combination ipilimumab/nivolumab showed a significant advantage in terms of PSF (HR=0.63; 90%CI 0.41 – 0.97) and ORR (28% vs. 9%) (VanderWalde et al., 2023). Grade 3 or higher irAEs occurred in 57% of patients treated in the combination arm and in 35% in the ipilimumab arm (VanderWalde et al., 2023). Of note, despite the small sample size, the subgroup analysis showed no significant advantage of the combination in patients who received more than 6

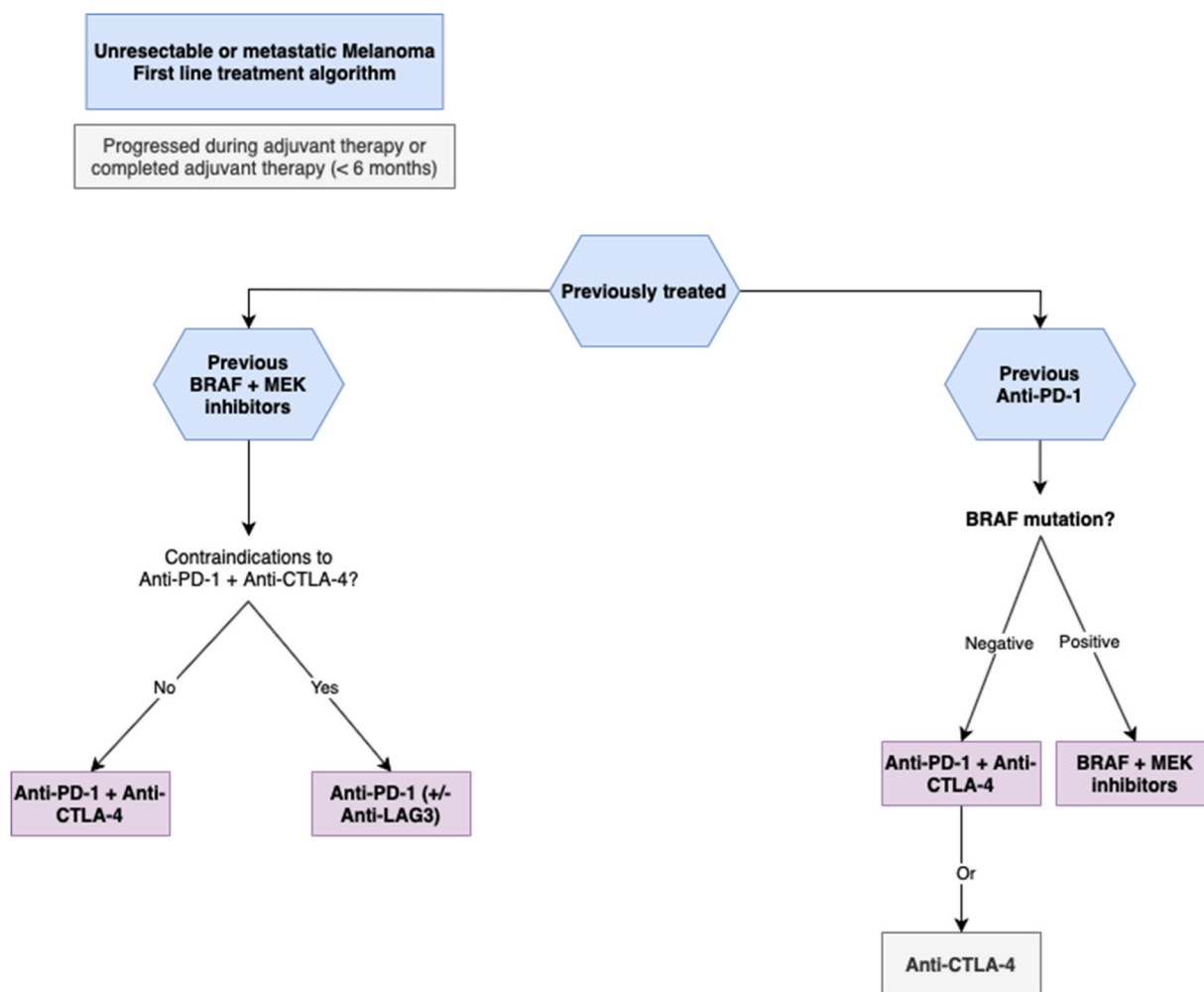


Fig. 3. Advanced Melanoma first-line treatment algorithm for patients progressing during or shortly after (before six months) completion of adjuvant therapy.

months of previous anti-PD(L)1 therapy and in those who received previous adjuvant therapy, favoring lesser pre-treated patients (VanderWalde et al., 2023). Second-line treatment strategies for advanced melanoma are being investigated, including the combination of nivolumab and relatlimab in patients who have progressed on anti-PD(L)1 therapies. In the RELATIVITY-020 trial, the efficacy and safety of nivolumab and relatlimab were assessed in patients with progression during or shortly after anti-PD(L)1 regimens. The ORRs were 12.0% and 9.2% in patient groups progressing respectively on one and on more than one anti-PD(L)1-containing regimens, and the median PFS ranged from 2.1 to 3.2 months. Importantly, the combination therapy showed durable clinical activity and a manageable safety profile in heavily pre-treated patients with advanced melanoma who had previously progressed on anti-PD(L)1 therapies (Ascierto et al., 2023).

Regarding patients who experienced a recurrence after adjuvant therapy with BRAF/MEK inhibitors, a multicentre retrospective study observed that among 85 patients with a median time to first recurrence of 18 months, transitioning to immunotherapy (specifically ipilimumab/nivolumab) demonstrated a response rate of approximately 60%, whereas a rechallenge with targeted therapy yielded a response rate of 25% (Bhave et al., 2021).

In summary, for patients experiencing recurrence during adjuvant therapy with anti-PD-1 or within six months of its discontinuation, it is recommended to initiate first-line treatment with anti-PD-1 plus anti-CTLA-4. In the case of patients with BRAF-mutant melanoma, especially those with a poor performance status or when a rapid response to immunotherapy is not expected or needed because of a high risk of early

death, first-line therapy with BRAF/MEK inhibitors could represent a viable alternative (see Figs. 1 and 2).

In case of treatment-naïve patients or patients who completed adjuvant anti-PD-1 more than six months earlier, the therapeutic decision should be based on prognostic assessments. The advantage gained from the ipilimumab/nivolumab combination, in fact, increases as prognostic factors worsen (Hodi et al., 2022). Particularly, for high LDH values, while the overall survival efficacy of encorafenib/binimetinib drastically decreases (Dummer et al., 2022), that of ipilimumab/nivolumab decreases while still maintaining a 40% survival rate at 5 years (with a difference of approximately 10% compared to nivolumab monotherapy) (Hodi et al., 2022). A similar trend has also been observed in the SECOMBIT study (Ascierto et al., 2022). PD-L1 also plays a prognostic role that can, in some cases, guide therapeutic choices. Although the CheckMate 067 study was not designed to directly compare the ipilimumab/nivolumab and nivolumab/placebo arms, a significant difference in overall survival (approximately 15%) is observed in the PD-L1 negative population (Hodi et al., 2022). This difference is not maintained in the PD-L1 positive group (Hodi et al., 2022). It is also worth noting that the efficacy of the ipilimumab/nivolumab combination has been tested in patients with a good performance status (Hodi et al., 2022). Therefore, in the presence of a poor performance status and BRAF-mutant melanoma, a combination of BRAF/MEK inhibitors would be preferable.

Another subgroup of patients is those with brain metastases. As discussed in Chapter 5, and as shown in Figs. 1 and 2, in cases of asymptomatic brain metastases, the recommended therapy is the

combination of ipilimumab/nivolumab, while in cases of symptomatic brain metastases, in the presence of the BRAF mutation, opting for a combination of BRAF/MEK inhibitors is recommended.

However, not all patients can be treated with an immunotherapy combination. In the case of fragile patients with comorbidities, autoimmune diseases that contraindicate immunotherapy combination, lack of psychosocial support, there are prognostic considerations that may lead us to choose anti-PD-1 monotherapy (see Figs. 1 and 2). Specifically, in cases of normal LDH, positive PD-L1, low tumor burden, or in the presence of variants like desmoplastic melanoma, these factors may encourage the use of anti-PD-1 monotherapy. If all these favorable prognostic characteristics are not present but greater clinical activity is required, the combination of anti-PD-1 plus anti-LAG-3 may be a valid alternative.

8. Challenges and future perspectives

One of the most clinically relevant current challenges is the management of patients who develop resistance to anti-PD-1 based treatment. Strategies to overcome resistance to anti-PD-(L)1 have been explored with the addition of a multikinase inhibitor to pembrolizumab. The phase II LEAP-004 study evaluated the combination of lenvatinib and pembrolizumab, in patients with melanoma that progressed on PD-1/L1 inhibitors. A total of 103 patients were treated, and the ORR was 21.4%, with complete responses observed in 2.9% and partial responses in 18.4% of patients. The median duration of response was 8.3 months. Notably, patients who had previously received anti-PD-1 plus anti-CTLA-4 therapy had an ORR of 33.3%. The median progression-free survival was 4.2 months, and the overall survival was 14.0 months. Treatment-related adverse events occurred in 45.6% of patients, with hypertension being the most common. The study results suggest that the combination of lenvatinib and pembrolizumab may be a future option for previously treated advanced melanoma patients (Arance et al., 2023). However, the phase 3 LEAP-003 trial was halted as this combination did not meet its primary endpoint of improved OS in the first-line setting ([67]).

Another potential salvage strategy was presented during the presidential symposium at the ESMO Congress 2022. This phase 3 trial aimed to assess the role of adoptive cell therapy with tumor-infiltrating lymphocytes (TILs) in the treatment of advanced melanoma. Patients with unresectable stage IIIC or IV melanoma were randomly assigned to receive either TIL therapy or ipilimumab. The infusion of TILs was preceded by lymphodepleting chemotherapy and followed by high-dose interleukin-2. Most of the included patients (90%) received previous systemic anti-PD-1 treatment. The study found that patients in the TIL group had a significantly longer median progression-free survival of 7.2 months compared to 3.1 months in the ipilimumab group. Specifically, the subgroup that derived the greatest benefit consisted of patients previously treated with anti-PD-1 for advanced disease (HR 0.39; 95%CI 0.26 - 0.61). The TIL group also showed a higher objective response rate of 49% compared to 21% in the ipilimumab group. Median overall survival was also longer in the TIL group (25.8 months) compared to the ipilimumab group (18.9 months). All patients in the TILs group experienced a grade ≥ 3 treatment-related adverse, primarily chemotherapy-related myelosuppression. These results suggest that TIL therapy may have a beneficial role in improving outcomes for patients with advanced melanoma (even in patients who have received prior treatment). However, it should be noted that patients must meet certain criteria for tolerating the high toxicity associated with this treatment. In fact, the patients enrolled in this trial had to be highly selected, especially considering the high response rate achieved by patients treated in the ipilimumab arm (approximately 20%) (Rohaani et al., 2022). Moreover, administration of TIL therapy requires specialized and well-equipped hospital facilities (Rohaani et al., 2022). Chimeric Antigen Receptor (CAR) T-cell therapy, while still in its early phase experimental stages, could be a promising approach in melanoma treatment

(Merhavi-Shoham et al., 2017). This innovative therapy involves the genetic modification of T cells to express CARs, enabling them to specifically target and attach to melanoma cells (Merhavi-Shoham et al., 2017). Current research and clinical trials focus on assessing the activity and safety of CAR T-cell therapy in melanoma, aiming to provide new treatment options for patients resistant to conventional therapies (Robbins et al., 2015). The development of this therapy also includes addressing challenges such as identifying specific tumour target antigens and avoiding autoimmune toxicities, emphasizing the need for tumour-specific antigens, and considering the therapy's technical feasibility and limitations related to antigen expression and HLA restriction (Merhavi-Shoham et al., 2017; Robbins et al., 2015).

Several ongoing efforts are currently underway to identify viable second-line treatment options, particularly for patients with BRAF wild-type melanoma (Table 1). Collectively, these studies contribute to expanding treatment options and addressing the unmet needs in second-line therapy for advanced melanoma, particularly in patients who have not responded to or have progressed on anti-PD-(L)1 immunotherapy.

A phase 3 study currently in progress (Table 1) is evaluating the efficacy of tebentafusp alone or in combination with pembrolizumab in HLA-A* 02:01-positive patients with advanced non-ocular melanoma (NCT05549297).

Furthermore, with the objective of overcoming primary or acquired resistance to PD-1 inhibitors, there is currently ongoing development of novel immunomodulatory drugs. IO102-IO103 is a potentially groundbreaking dual-antigen, immune-modulatory therapy designed to target the key immune resistance pathways mediated by IDO and PD-L1. Pre-clinical and clinical studies have demonstrated the ability of IO102 and IO103 to activate specific T cells recognizing these checkpoint molecules and modulate immune regulation. Preclinical models have also shown a synergistic anti-tumor response when both IDO and PD-L1 are targeted simultaneously in the tumor microenvironment. Combining the distinctive mechanisms of action of IO102-IO103 and anti-PD-1 antibodies may potentially result in a further synergistic effect. Promising results have been observed in a previous Phase 1/2 trial investigating the combination of IO102-IO103 and nivolumab in anti-PD-1-naïve metastatic melanoma patients, with high overall response rates, prolonged progression-free survival, and a manageable safety profile. Building upon these findings, a Phase 3 multicenter, open-label, randomized trial is currently ongoing to evaluate the efficacy and safety of IO102-IO103 plus pembrolizumab versus pembrolizumab alone in untreated, unresectable, or metastatic melanoma patients (NCT05155254).

9. Conclusions

In conclusion, the treatment landscape for advanced melanoma has witnessed remarkable progress, driven by the introduction of immune-checkpoint inhibitors and targeted therapies. This review has provided a simplified overview of first-line treatment strategies for BRAF-mutant and wild-type melanoma, along with discussing important clinical trials conducted in patients with brain metastases and updates on the management of rare histologies such as uveal and mucosal melanoma. The findings from this research offer valuable insights that can guide clinicians in making informed decisions and tailoring therapies to individual patients (refer to Fig. 1 for different clinical scenarios, and Fig. 2 for the proposed treatment decision algorithm). Notably, the challenge of managing patients who develop resistance to anti-PD-1 treatment remains a significant clinical concern. The limited availability of therapeutic salvage strategies underscores the need for further research and innovative approaches to address this issue.

CRedit authorship contribution statement

Conceptualization, A.B.; methodology, A.B.; validation, F.S., R.M. and E.T.T.; writing—original draft preparation, A.B., E.C., M.F., R.G., G.

Table 1

Active phase 3 clinical trials recruiting patients with advanced melanoma.

NCT Number	Title	Status	Interventions	Primary Endpoint	Enrollment	Start Date
NCT05789043	Camrelizumab in Combination With Apatinib and Temozolomide as First-line Treatment in Advanced Acral Melanoma	Not yet recruiting	Experimental: Camrelizumab + Apatinib + TMZ Active Comparator: Camrelizumab + Apatinib Active Comparator: Camrelizumab	PFS	140	March 20, 2023
NCT05732805	A Clinical Study of BCD-217 (Nurulimab + Prolgolimab) Followed by Anti-PD-1 Compared to Anti-PD-1 Monotherapy as First-Line Treatment in Subjects With Unresectable/Metastatic Melanoma (OCTAVA)	Recruiting	Experimental: Nurulimab + Prolgolimab Active Comparator: Prolgolimab	PFS	270	August 2, 2022
NCT05727904	Study to Investigate Lifileucel Regimen Plus Pembrolizumab Compared With Pembrolizumab Alone in Participants With Untreated Advanced Melanoma.	Recruiting	Experimental: Lifileucel plus Pembrolizumab Active Comparator: Pembrolizumab with Optional Crossover Period	ORR, PFS	670	March 30, 2023
NCT05647954	A Study of HX008 Plus Transcatheter Arterial Chemoembolization (TACE) in the First-Line Treatment of Subjects With Stage IV (M1c) Melanoma That is Metastatic to the Liver	Not yet recruiting	Experimental: HX008 Plus TACE Active Comparator: Temozolomide Plus TACE Active Comparator: Pembrolizumab	PFS	350	December 31, 2022
NCT05625399	A Study of Subcutaneous Nivolumab + Relatlimab Fixed-dose Combination (FDC) in Previously Untreated Metastatic or Unresectable Melanoma (RELATIVITY-127)	Recruiting	Experimental: Nivolumab + Relatlimab FDC SC Active Comparator: Nivolumab + Relatlimab FDC IV	Time-averaged serum concentration over 28 days after the first dose (Cavgd28) of Nivolumab	814	March 6, 2023
NCT05549297	Tebentafusp Regimen Versus Investigator's Choice in Previously Treated Advanced Melanoma (TEBE-AM)	Recruiting	Experimental: Tebentafusp Experimental: Tebentafusp with Pembrolizumab Active Comparator: Investigators Choice	Adverse Events and Severe Adverse Events	460	December 19, 2022
NCT05522660	Immunotherapy or Targeted Therapy With or Without Stereotactic Radiosurgery for Patients With Brain Metastases From Melanoma or Non-small Cell Lung Cancer (USZ-STRIKE)	Recruiting	Experimental: standard systemic treatment with stereotactic radiosurgery (SRS) Active Comparator: standard systemic treatment without stereotactic radiosurgery	CNS-specific PFS	190	November 30, 2022
NCT05352672	Clinical Study of Fianlimab in Combination With Cemiplimab in Adolescent and Adult Patients With Previously Untreated Unresectable Locally Advanced or Metastatic Melanoma	Recruiting	Experimental A: Fianlimab + Cemiplimab dose 1 Experimental A1: Fianlimab + Cemiplimab dose 2 Experimental B: Pembrolizumab + placebo Experimental C: Cemiplimab + placebo	PFS	1590	July 14, 2022
NCT05155254	IO102-IO103 in Combination With Pembrolizumab Versus Pembrolizumab Alone in Advanced Melanoma (IOB-013 / KN-D18)	Recruiting	Experimental: IO102-IO103 + Pembrolizumab Active Comparator: Pembrolizumab	PFS	300	December 21, 2021
NCT05022901	An Open-Label Expanded Access Study of the Melphalan/Hepatic Delivery System (HDS) in Patients With Hepatic Dominant Ocular Melanoma	Recruiting	Experimental: Melphalan (3 mg/kg IBW) with Hepatic Delivery System (HDS)	Adverse Events	30	June 10, 2022
NCT04674683	Study Comparing Investigational Drug HBI-8000 Combined With Nivolumab vs. Nivolumab in Patients With Advanced Melanoma	Recruiting	Experimental: HBI-8000 30 mg oral BIW + Nivolumab IV Active Comparator: Placebo oral BIW + Nivolumab IV	ORR	480	August 12, 2021
NCT04657991	A Clinical Trial of Three Study Medicines (Encorafenib, Binimetinib, and Pembrolizumab) in Patients With Advanced or Metastatic Melanoma (STARBOARD)	Recruiting	Experimental: Encorafenib and Binimetinib in combination with Pembrolizumab Active Comparator: Pembrolizumab	PFS	624	January 15, 2021
NCT02821013	Duration of Anti-PD-1 Therapy in Metastatic Melanoma (STOP-GAP)	Recruiting	Active Comparator Arm 1: Intermittent PD-1 Inhibitor therapy Active Comparator Arm 2: Continuous PD-1 Inhibitor therapy	ORR	614	July 4, 2016
NCT02339571	A Phase II/III Trial of Nivolumab, Ipilimumab, and GM-CSF in Patients With Advanced Melanoma	Recruiting	Experimental Arm A: Nivolumab, Ipilimumab, Sargramostim Experimental Arm B: Nivolumab, Ipilimumab	OS	600	September 10, 2015

ORR: overall response rate; OS: overall survival; PFS: progression-free survival.

Table 2
Selected key phase 3 studies in advanced melanoma.

Study name	Author, year	Intervention	Number of patients	ORR n (%)	mOS [months] (95% CI)	TRAEs ≥ G3 n (%)
CheckMate 067	Wolchok JD, 2022; Larkin J, 2019; Hodi FS, 2018; Wolchok JD, 2017; Larkin J, 2015	Nivolumab 1 mg/kg plus Ipilimumab 3 mg/kg	314	181 (57.6)	72.1 (38.2 – NR)	186 (59)
		Nivolumab plus Placebo	316	138 (43.7)	36.9 (28.2 – 58.7)	73 (23)
		Ipilimumab plus Placebo	315	60 (19)	19.9 (16.8 – 24.6)	86 (28)
		Relatlimab plus Nivolumab	355	43.1%	NR (34.2–NR)	79 (22.2)
RELATIVITY-047	Tawbi HA, 2022; Long GV, 2022	Nivolumab plus Placebo	359	32.6%	34.1 (25.2–NR)	42 (11.7)
		Spartalizumab plus Dabrafenib plus Trametinib	267	183 (69)	NR (30.6–NR)	146 (55)
		Dabrafenib plus Trametinib plus Placebo	265	170 (64)	NR (28.3–NR)	88 (33)
		Atezolizumab plus Vemurafenib plus Cobimetinib	256	66.3%	39.0	79%
IMspire150	Gutzmer R, 2020; McArthur G.A., 2022	Vemurafenib plus Cobimetinib plus Placebo	258	65%	25.8	73%
		Nivolumab	210	84 (40)	37.3 (25.4–51.6)	33 (16)
		Dacarbazine	208	29 (14)	11.2 (9.6–13.0)	36 (18)
		Pembrolizumab 10 mg/kg Q21d	185	36%	32.7 (24.5–41.6)	17%
CheckMate 066	Robert C, 2020; Ascierto PA, 2019; Robert C, 2014	Pembrolizumab 10 mg/kg Q14d	183	37%		17%
		Ipilimumab	181	13%	16 (13.3–22.0)	20%
		Encorafenib plus Binimetinib	192	121 (63)	33.6 (24.4–39.2)	131 (68.2)
		Encorafenib	194	98 (51)	23.5 (19.6–33.6)	130 (67.7)
KEYNOTE-006	Robert C, 2019; Schachter J, 2017; Robert C, 2015	Vemurafenib	191	77 (40)	16.9 (14.0–24.5)	122 (65.6)
		Dabrafenib plus Trametinib	211	140 (67)	25.1 (19.2–NR)	67 (32)
		Dabrafenib plus Placebo	212	108 (51)	18.7 (15.2–23.7)	66 (31.5)
		Dabrafenib plus Trametinib	352	64%	NR	173 (49%)
COLUMBUS	Ascierto PA, 2020; Dummer R, 2018	Vemurafenib	352	51%	17.2	206 (59)
		Vemurafenib plus Cobimetinib	247	172 (70)	22.5 (20.3–28.8)	147 (60)
		Vemurafenib plus Placebo	248	123 (50)	17.4 (15.0–19.8)	128 (52)
COMBI-d	Robert C, 2019; Long GV, 2017; Long GV, 2015; Long GV, 2014					
COMBI-v	Grob JJ, 2015; Robert C, 2014					
coBRIM	Ascierto PA, 2021; Ascierto PA, 2016; Larkin J, 2014					

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Declaration of Competing Interest

FS has reported Honoraria for presentations or lectures from Sanofi Genzyme, Roche, BMS, Novartis, Merk, Sun Pharma, MSD, Pierre Fabre and for advisory board for Novartis, Philogen, SunPharma, MSD. The other authors have no conflict of interest to declare.

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